

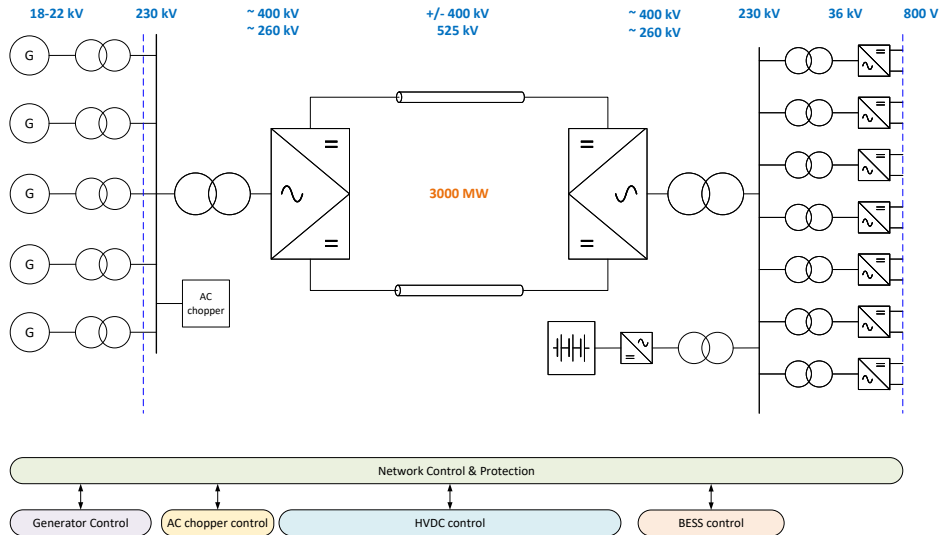
AMERICAN TERAWATT

System Configuration

RTEi & American Terawatt

American Terawatt | July 13, 2026

System configuration



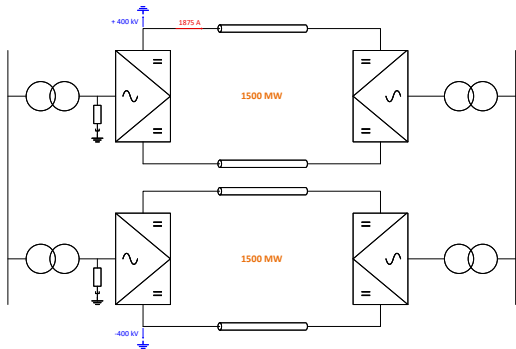
System configuration

- **Power** - Projects with $1000MW$ and $3000MW$ power transmission requirements
- **DC line** - 130 -200 miles, with priority to OHL, but cable sections can be implemented to accelerate the permission process.
- **Generation** - Geothermal power plants is the main generation source. Additional generation sources might be implemented once the transmission system is established.
- **Load** - The main load are data centers. The interconnection voltage to the data centers to be defined ($36kV$ or $800V$).
- **BESS** - Battery storage is assumed for backup and for datacenter load flicker compensation.
- **AC Chopper** - For fault ride through, installation of the AC chopper might be required.

System configuration

- Islanded operation with no interconnection to the AC grid
- Generation is build out in $300MW$ steps and is defined by the water availability
⇒ Staged buildout of the transmission line is beneficial
- High reliability requirements by datacenters
- Execution time is critical
- Standardized hardware design approach to reduce hardware lead times and configuration variations
- Building up C&P capabilities to reduce risk and execution times in project execution

HVDC Configuration - Dual SMP

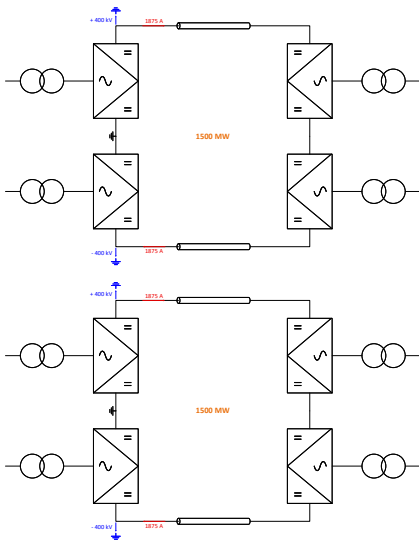


- **Topology** - Dual Symmetrical Monopole with $\pm 400kV$
- **DC Line** - DC cable and OHL can be utilized with the given current rating
- **Stages** - Two $1500MW$ stages
- **Flexibility** - Design can be used for configurations with $1000MW$ rating



- 6

HVDC Configuration - Rigid BIP



- **Topology** - Dual Rigid Bipole with $\pm 400kV$ and **3ph** transformer units
- **DC Line** - DC cable and OHL can be utilized with the given current rating
- **Stages** - 750MW stages for converter and 1500MW stages for DC line
- **Flexibility** - Limited flexibility as asymmetrical configuration is not economical for most requirements

Working Packages

■ DC line

- Local company is working on DC route definition and DC line design

■ Network Control & Protection

- Operation concept definition
- Offline studies for concept verification
- Development of models for performance design and verification
- Definition of performance verification tests and evaluation criteria
- Control hardware definition

■ HVDC concept evaluation

- Technical evaluation of HVDC concepts
- Commercial evaluation of HVDC concepts
- OEMs should be approached with only one concept

Working Packages

■ Project execution

- Converter specification to receive a binding offer (FEED phase avoided to reduce the execution time)
- Supplier evaluation and selection (it is expected that suppliers from Asia might be more suitable for our approach)
- Testlab buildout for real time simulations
- Project execution support (Owners Engineer)